

## Linux System Programming Lab Test

Duration: 45 Min;

Marks:10M

Write 2 programs (parent.c, children.c) which can run in parallel and communicate in a synchronised way as below:

In parent.c, mywait() function is to be implemented with below functionality:

In children.c, myexit() function is to be implemented with below functionality:

| parent.c                                                                                                                                                                                                                                                       | children.c                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <pre>main() {     mywait(); }  void mywait(void) {     keep reading data from a file abc, until it gets a 1     from the file. Keeps printing the data that it has     read (read one character every time and print     until a 1 is read from file). }</pre> | <pre>main() {     myexit(); }  void myexit(void) {     writes a 1 to the abc on execution. }</pre> |

Create separate binary files for parent and children don't use fork() and execve() system calls. Before execute above programs current folder abc is mandatory and initialize to 0.

Hint: abc file it can store only one byte data that is 0 or 1.

mywait() function from parent.c will keep reading and printing from abc file, and whenever myexit() function writes a 1 to the abc file, children.c program will stop.